VANISHING AGAINST SKYSCRAPER SHEAVES SPREADS TO NEIGHBOURHOODS

THE STACKS PROJECT AUTHORS

Allowed standard prerequisites (editorial).

- Cut complex in two. more-algebra-lemma-cut-complex-in-two (cross-chapter, named). Engine of part (5).
- Point duality. lemma-duality-at-point L26479 (carded this batch). Feeds part (1) via Matlis duality.
- Perfect duality. cohomology-lemma-dual-perfect-complex (cross-chapter, named). Feeds part (3) and the double-dual step of part (4).

Source reference. Original reference 08DQ: focused-seeds/geometry/S9-eqv-orthogonal-point-sheaf/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0G02: sources/stacks-equiv/pinned-original-source.tex, source lines 2748–2748. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 068U: focused-seeds/geometry/S9-eqv-orthogonal-point-sheaf/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let k be a field. Let X be a scheme of finite type over k which is regular. Let $x \in X$ be a closed point and denote $\mathcal{O}_x$ the skyscraper sheaf at x with value $\kappa(x)$. Let K in $D_{perf}(\mathcal{O}_X)$.*

(1) *If $\mathrm{Ext}^i_X(\mathcal{O}_x, K) = 0$ then there exists an open neighbourhood U of x such that $H^{i-d_x}(K)|_U = 0$ where $d_x = \dim(\mathcal{O}_{X,x})$.*
(2) *If $\mathrm{Hom}_X(\mathcal{O}_x, K[i]) = 0$ for all $i \in \mathbf{Z}$, then K is zero in an open neighbourhood of x.*
(3) *If $\mathrm{Ext}^i_X(K, \mathcal{O}_x) = 0$ then there exists an open neighbourhood U of x such that $H^i(K^\vee)|_U = 0$.*
(4) *If $\mathrm{Hom}_X(K, \mathcal{O}_x[i]) = 0$ for all $i \in \mathbf{Z}$, then K is zero in an open neighbourhood of x.*
(5) *If $H^i(X, K \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{O}_x) = 0$ then there exists an open neighbourhood U of x such that $H^i(K)|_U = 0$.*

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(6) *If* $H^i(X, K \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{O}_x) = 0$ *for* $i \in \mathbf{Z}$ *then* K *is zero in an open neighbourhood of* x*.*

Proof. Observe that $H^i(X, K \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{O}_x)$ is equal to $K_x \otimes^{\mathbf{L}}_{\mathcal{O}_{X,x}} \kappa(x)$. Hence part (5) follows from More on Algebra, Lemma 068U. Part (6) follows from part (5). Part (1) follows from part (5), Lemma 0G02, and the fact that the Matlis dual of $\kappa(x)$ is $\kappa(x)$. Part (2) follows from part (1). Part (3) follows from part (5) and the fact that $\mathrm{Ext}^i(K, \mathcal{O}_x) = H^i(X, K^\vee \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{O}_x)$ by Cohomology, Lemma 08DQ. Part (4) follows from part (3) and the fact that $K \cong (K^\vee)^\vee$ by the lemma just cited. □